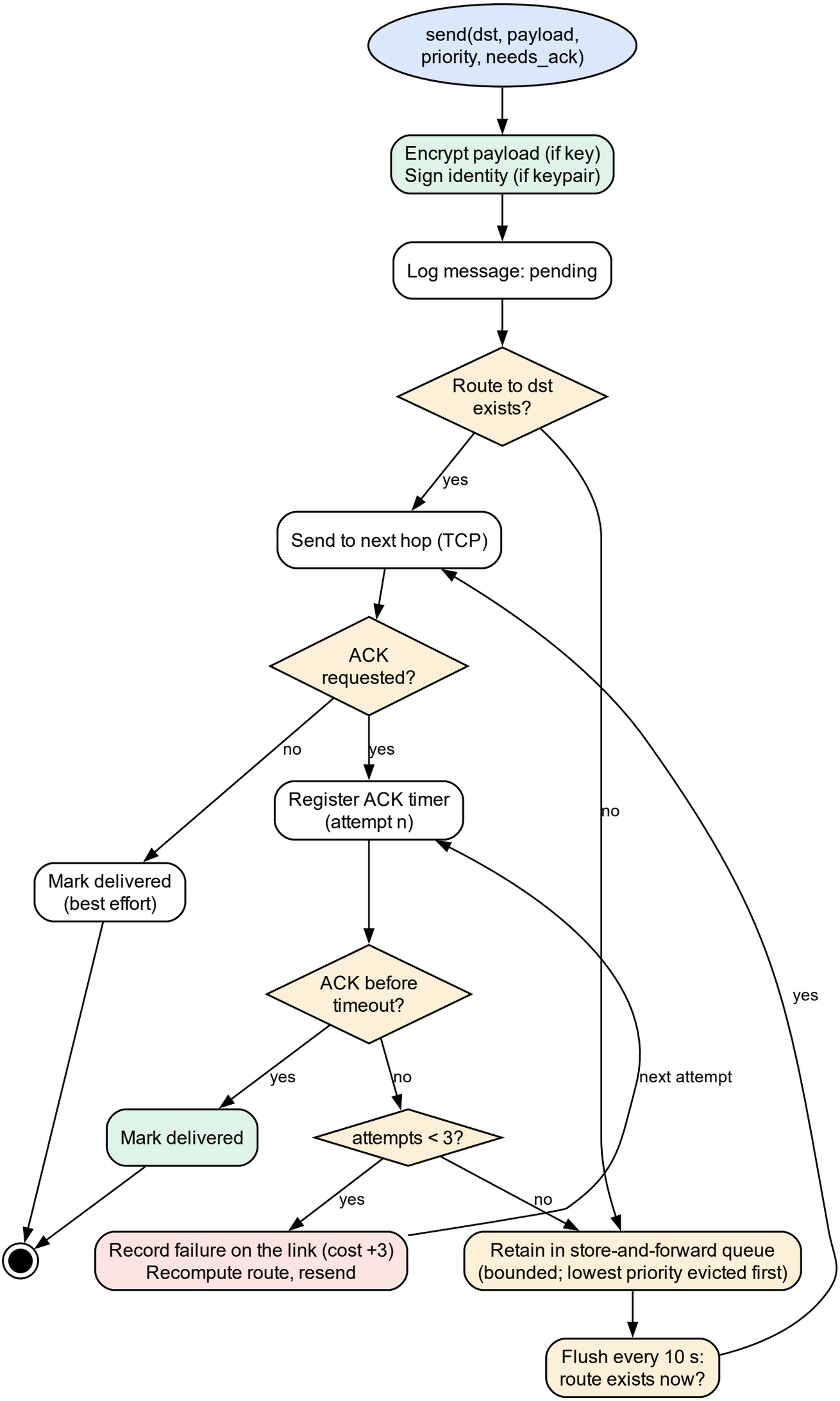
send(dst, payload, priority, needs_ack)
Encrypt payload (if key) Sign identity (if keypair)
Log message: pending
Route to dst exists?
yes
no
Send to next hop (TCP)
ACK requested?
no
yes
Register ACK timer (attempt n)
Mark delivered (best effort)
ACK before timeout?
yes
no
Mark delivered
attempts < 3?
yes
no
next attempt
Record failure on the link (cost +3) Recompute route, resend
Retain in store-and-forward queue (bounded; lowest priority evicted first)
Flush every 10 s: route exists now?
yes